Flight Dynamics & Control

Tutorial 1

1. Obtain the eigenvalues and eigenvectors of the following matrices

   (a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

   (b) $\mathbf{A} = \begin{bmatrix} 1 & -5 \\ 2 & -1 \end{bmatrix}$

   (c) $\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

   (d) $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(a) $\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$

$\therefore (1-\lambda)(3-\lambda) - 8 = 0$

$\therefore \lambda^2 - 4\lambda - 5 = (\lambda+1)(\lambda-5) = 0$

$\therefore \lambda_1 = -1$ and $\lambda_2 = 5$

for $\lambda_1 = -1$,

$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \xi_1 = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

for $\lambda_2 = 5$

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \xi_2 = \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

(b) $\begin{vmatrix} 1-\lambda & -5 \\ 2 & -1-\lambda \end{vmatrix} = 0 \qquad \therefore -(1-\lambda)(1+\lambda) + 10 = 0$

$\therefore \lambda^2 + 9 = 0 \qquad \therefore \lambda_1 = 3i,\ \lambda_2 = -3i$

for $\lambda_1 = 3i$,

$\begin{bmatrix} 1-3i & -5 \\ 2 & -1-3i \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ \frac{1-3i}{5} \end{bmatrix}$

for $\lambda_2 = -3i$,

$\begin{bmatrix} 1+3i & -5 \\ 2 & -1+3i \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ \frac{1+3i}{5} \end{bmatrix}$

(c) $\begin{vmatrix} -\lambda & 0 & 0 \\ 1 & -\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} = 0$

$\therefore -\lambda[-\lambda(1-\lambda)] + 0 + 0 = \lambda^2(1-\lambda) = 0$

$\therefore \lambda_1 = 1,\ \lambda_{2,3} = 0$

for $\lambda_1 = 1$

$$\begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xi_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xi_1 = 0 \quad \therefore \xi_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

for $\lambda_{2,3} = 0$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \xi_2 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xi_2 = 0 \quad \therefore \xi_{2,3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

(d) $$\begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & -\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \xi_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\rightarrow \xi_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\therefore -\lambda(1-\lambda)^2 = 0 \quad \therefore \lambda_1 = 0$ or $\lambda_{2,3} = 1$

for $\lambda_1 = 0$,

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xi_1 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xi_1 = 0 \quad \therefore \xi_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

for $\lambda_{2,3} = 1$,

$$\begin{bmatrix} 0 & 0 & 2 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xi_2 = 0 \quad \therefore \xi_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 2 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xi_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \therefore \xi_3 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

2. For any given initial condition, obtain the general solution for the following systems or differential equations.

(a) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(b) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(c) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(d) $\ddot{x} + \dot{x} - 6x = 0$

(e) $\dddot{x} - 3\ddot{x} + 3\dot{x} - x = 0$

(a) $x' = x^* e^{\lambda t}$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \xi_1 e^{\lambda_1 t} + k_2 \xi_2 e^{\lambda_2 t}$$

$$\therefore \begin{vmatrix} -\lambda & 1 \\ -6 & -5-\lambda \end{vmatrix} = 0 \qquad \therefore \lambda(5+\lambda) + 6 = 0$$

$$\therefore \lambda^2 + 5\lambda + 6 = (\lambda+2)(\lambda+3) = 0$$

$$\therefore \lambda_1 = -2, \ \lambda_2 = -3$$

for $\lambda_1 = -2$,

$$\begin{bmatrix} 2 & 1 \\ -6 & -3 \end{bmatrix} \xi_1 = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

for $\lambda_2 = -3$

$$\begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} \xi_2 = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-2t} + k_2 \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{-3t}$$

$$\therefore \begin{cases} x_1(0) = k_1 + k_2 \\ x_2(0) = -2k_1 - 3k_2 \end{cases}$$

(b) $X' = x^* e^{\lambda t}$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \xi_1 e^{\lambda_1 t} + k_2 \xi_2 e^{\lambda_2 t}$$

$$\begin{vmatrix} 4-\lambda & 1 \\ -1 & 2-\lambda \end{vmatrix} = 0 \qquad \therefore (4-\lambda)(2-\lambda) + 1 = 0$$

$$\therefore \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0$$

$$\therefore \lambda_1 = \lambda_2 = 3$$

for $\lambda_1 = \lambda_2 = 3$,

$$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \xi_1 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \xi_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t} + k_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{3t}$$

$$\therefore \begin{cases} x_1(0) = k_1 + k_2 \\ x_2(0) = -k_1 \end{cases}$$

(c) $x' = x^* e^{\lambda t}$

$$\begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = 0 \qquad \therefore \lambda^2 + 1 = 0 \qquad \therefore \lambda = \pm i$$

for $\lambda_1 = i$,

$$\begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \xi_1 = \begin{bmatrix} -i & 1 \\ 0 & 0 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

for $\lambda_2 = -i$,

$$\begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix} \xi_2 = \begin{bmatrix} i & 1 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{it} + k_2 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{-it}$$

$$\therefore \begin{cases} x_1(0) = k_1 + k_2 \\ x_2(0) = ik_1 - ik_2 \end{cases}$$

(d) $\ddot{x} + \dot{x} - 6x = 0$

$$\therefore \begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 6 & -1 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$$

$$\therefore \begin{vmatrix} -\lambda & 1 \\ 6 & -1-\lambda \end{vmatrix} = 0$$

$$\therefore \lambda(1+\lambda) - 6 = \lambda^2 + \lambda - 6 = (\lambda-2)(\lambda+3) = 0$$

$$\therefore \lambda_1 = 2, \lambda_2 = -3$$

for $\lambda_1 = 2$,

$$\begin{bmatrix} -2 & 1 \\ 6 & -3 \end{bmatrix} \xi_1 = \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

for $\lambda_2 = -3$,

$$\begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix} \xi_2 = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + k_2 \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{-3t}$$

$$\therefore \begin{cases} x_1(0) = k_1 + k_2 \\ x_2(0) = 2k_1 - 3k_2 \end{cases}$$

(e) $\dddot{x} - 3\ddot{x} + 3\dot{x} - x = 0$

$$\therefore \dot{X} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix}$$

$$\therefore \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & -3 & 3-\lambda \end{vmatrix} = 0$$

$$\therefore 0 - \lambda[-\lambda(3-\lambda)] + (3\lambda - 1) = -\lambda^3 + 3\lambda^2 + 3\lambda - 1 = 0$$

$$\therefore (\lambda - 1)^3 = 0 \qquad \therefore \lambda_1 = \lambda_2 = \lambda_3 = 1$$

for $\lambda_{1,2,3} = 1$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & -3 & 2 \end{bmatrix} \xi_{1,2,3} = 0 \qquad \therefore \xi_{1,2,3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + k_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + k_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t$$

$$\therefore \begin{cases} x_1(0) = k_1 + k_2 + k_3 \\ x_2(0) = k_1 + k_2 + k_3 \\ x_3(0) = k_1 + k_2 + k_3 \end{cases}$$

3. Linearise the following system about each of its equilibria and determine the local nature of the trajectories at these points.

$$\dot{x}_1 = x_1 x_2 + x_1^3$$

$$\dot{x}_2 = x_1 + x_2^2$$

Hence deduce the global phase plane picture

$$\dot{x} = f(x, \delta) = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_1 x_2 + x_1^3 \\ x_1 + x_2^2 \end{bmatrix}$$

$$\therefore J_{(x,\delta)} = \begin{bmatrix} x_2 + 3x_1^2 & x_1 \\ 1 & 2x_2 \end{bmatrix}$$

$$\therefore \dot{x}_0 = J(x_0, \delta) = 0$$

$$\therefore \begin{cases} x_1 x_2 + x_1^3 = 0 \\ x_1 + x_2^2 = 0 \end{cases}$$

$$\therefore x_1 = -x_2^2$$

$$\therefore -x_2^3 - x_2^6 = -x_2^3(1 + x_2^3) = 0$$

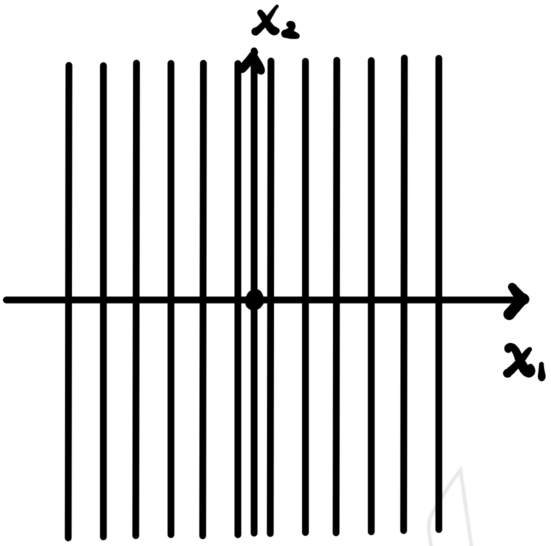
$$\therefore \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases} \quad \text{and} \quad \begin{cases} x_1 = -1 \\ x_2 = -1 \end{cases}$$

for $x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$J_{(x_0,\delta)} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$\therefore \begin{vmatrix} -\lambda & 0 \\ 1 & -\lambda \end{vmatrix} = 0 \quad \therefore \lambda^2 = 0 \quad \therefore \lambda_1 = 0$

$\therefore \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \xi_1 = 0 \quad \therefore \xi_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$



neutral stable

for $x_0 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

$J_{(x_0, \delta)} = \begin{bmatrix} 2 & -1 \\ 1 & -2 \end{bmatrix}$

$\therefore \begin{vmatrix} 2-\lambda & -1 \\ 1 & -2-\lambda \end{vmatrix} = 0$

$\therefore -(2-\lambda)(2+\lambda) + 1 = 0$

$\therefore \lambda^2 - 4 + 1 = \lambda^2 - 3 = 0 \quad \therefore \lambda_1 = \sqrt{3},\ \lambda_2 = -\sqrt{3}$

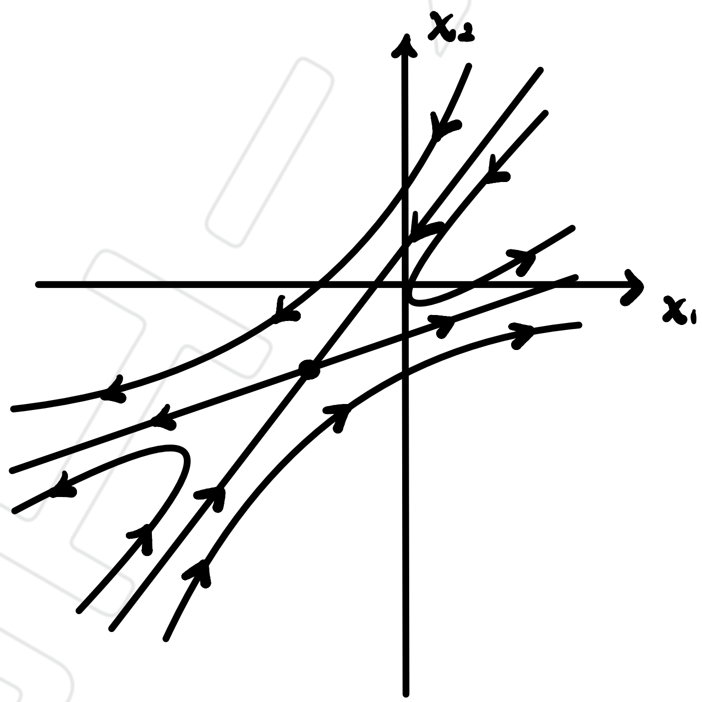
for $\lambda_1 = \sqrt{3}$,

$$\begin{bmatrix} 2-\sqrt{3} & -1 \\ 1 & -2-\sqrt{3} \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ 2-\sqrt{3} \end{bmatrix}$$

for $\lambda_2 = -\sqrt{3}$

$$\begin{bmatrix} 2+\sqrt{3} & -1 \\ 1 & -2+\sqrt{3} \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ 2+\sqrt{3} \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 2-\sqrt{3} \end{bmatrix} e^{\sqrt{3}t} + k_2 \begin{bmatrix} 1 \\ 2+\sqrt{3} \end{bmatrix} e^{-\sqrt{3}t}$$



4. Study, about the origin, the solution of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ for the following matrices

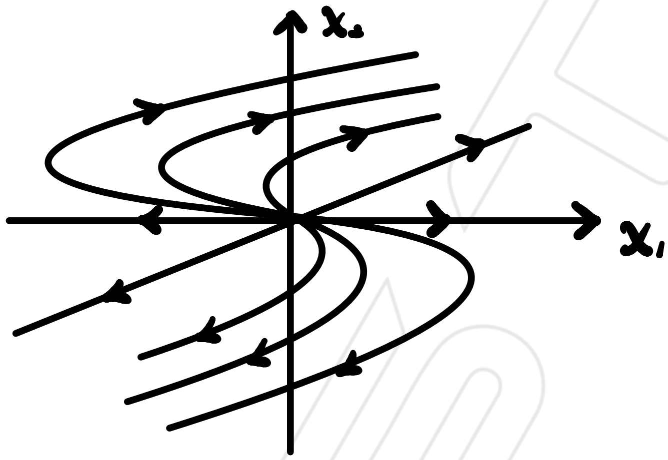
(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(d) $\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$

(e) $\mathbf{A} = \begin{bmatrix} -3 & -8 \\ 4 & 9 \end{bmatrix}$



(a) $\begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix} = 0 \qquad \therefore (1-\lambda)(2-\lambda) = 0$

$\therefore \lambda_1 = 1, \ \lambda_2 = 2$

for $\lambda_1 = 1$

$\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

for $\lambda_2 = 2$

$\begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix}$

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{t} + k_2 \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} e^{2t}$

(b) $\begin{vmatrix} -1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0$

$\therefore -(1+\lambda)(1-\lambda) = 0 \qquad \therefore \lambda_1 = 1,\ \lambda_2 = -1$

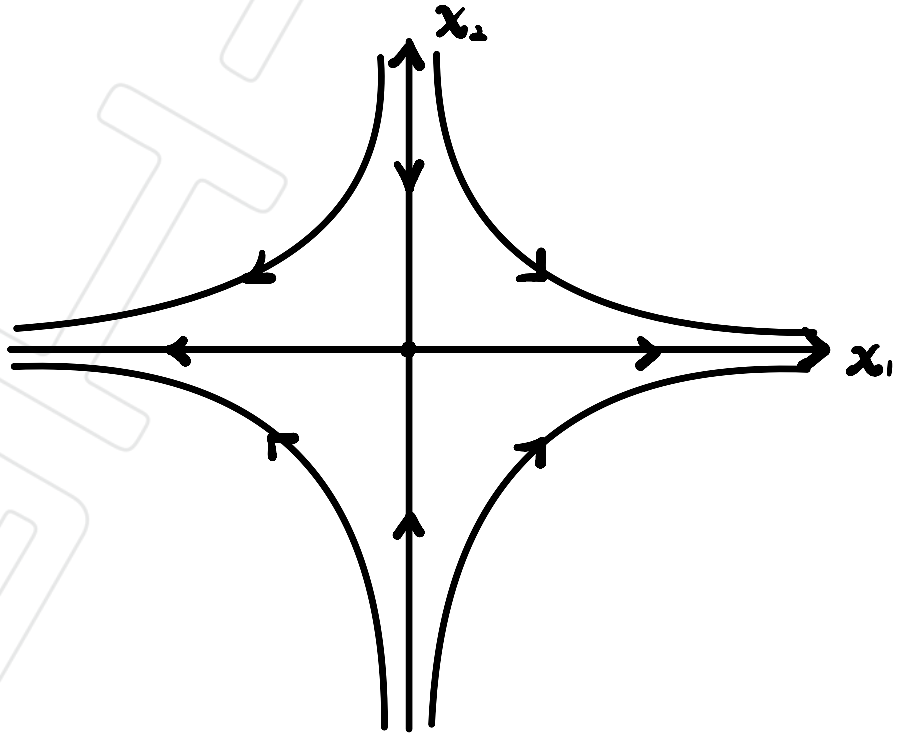
for $\lambda_1 = 1$,

$\begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

for $\lambda_2 = -1$,

$\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{t} + k_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t}$



(c) $\begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0$

$\therefore \lambda^2 - 1 = 0 \qquad \therefore \lambda_1 = 1,\ \lambda_2 = -1$

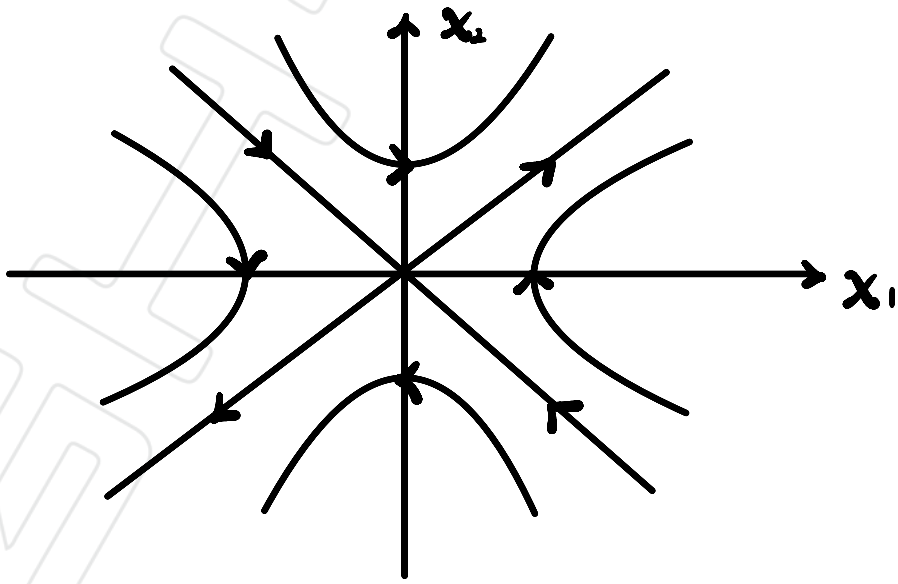
for $\lambda_1 = 1$,

$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

for $\lambda_2 = -1$,

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{\lambda t} + k_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-\lambda t}$



(d) $\begin{vmatrix} 1-\lambda & -2 \\ 0 & -1-\lambda \end{vmatrix} = 0$

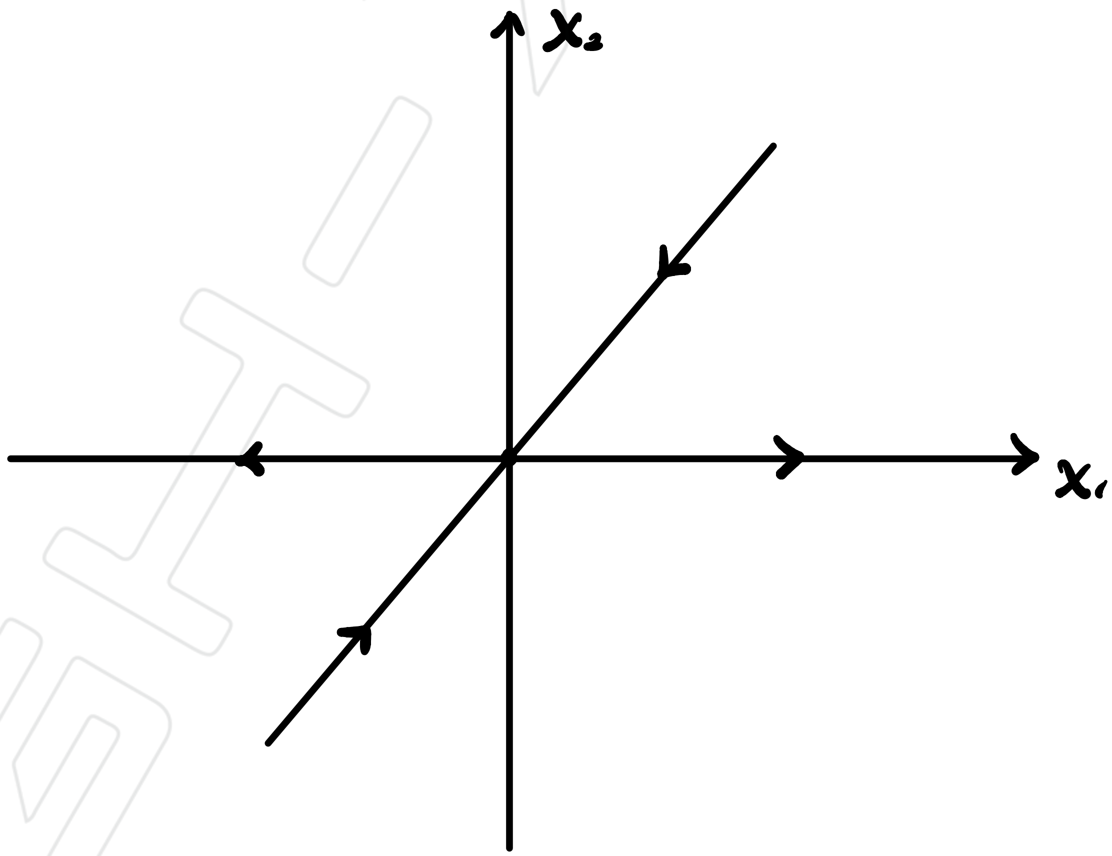
$\therefore -(1-\lambda)(1+\lambda) = 0 \quad \therefore \lambda^2 - 1 = 0 \quad \therefore \lambda_1 = 1,\ \lambda_2 = -1$

for $\lambda_1 = 1$,

$\begin{bmatrix} 0 & -2 \\ 0 & -2 \end{bmatrix} \xi_1 = 0 \quad \therefore \xi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

for $\lambda_2 = -1$,

$\begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \quad \therefore \xi_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$



(e) $\begin{vmatrix} -3-\lambda & -8 \\ 4 & 9-\lambda \end{vmatrix} = 0$

$\therefore \lambda^2 - 6\lambda - 27 + 32 = \lambda^2 - 6\lambda + 5 = (\lambda - 1)(\lambda - 5) = 0$

$\therefore \lambda_1 = 1,\ \lambda_2 = 5$

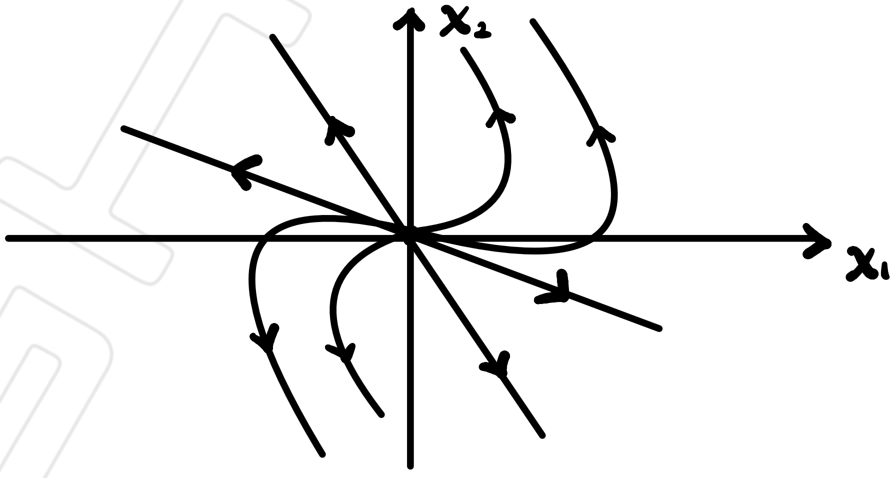
for $\lambda_1 = 1$,

$\begin{bmatrix} -4 & -8 \\ 4 & 8 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$

for $\lambda_2 = 5$.

$\begin{bmatrix} -8 & -8 \\ 4 & 4 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix} e^{t} + k_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{5t}$



5. Plot the state space structure for the system

$$\dot{x} = y$$

$$\dot{y} = \cos x$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} y \\ \cos x \end{bmatrix}$$

$$\therefore J_{(x,\delta)} = \begin{bmatrix} 0 & 1 \\ -\sin x & 0 \end{bmatrix}$$

At singular point,

$$\dot{X}_0 = f(X_0, \delta) = 0$$

$$\therefore \begin{bmatrix} y \\ \cos x \end{bmatrix} = 0 \qquad \therefore \begin{bmatrix} y \\ x \end{bmatrix} = \begin{bmatrix} 0 \\ \pm\frac{\pi}{2} + 2n\pi \end{bmatrix} \ (n \in \mathbb{R})$$

$$\text{for } \begin{bmatrix} y \\ x \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{\pi}{2} + 2n\pi \end{bmatrix} \ (n \in \mathbb{R})$$

$$\therefore J_{(x_0,\delta)} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \therefore \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 + 1 = 0 \qquad \therefore \lambda_1 = i, \ \lambda_2 = -i$$

for $\lambda_1 = i$,

$$\therefore \begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

for $\lambda_2 = -i$,

$$\therefore \begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{it} + k_2 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{-it}$$

for $\begin{bmatrix} y \\ x \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{\pi}{2} \pm 2n\pi \end{bmatrix} \ (n \in \mathbb{R})$

$$\therefore J_{(x_0, y_0)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\therefore \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \qquad \therefore \lambda^2 - 1 = 0$$

$$\therefore \lambda_1 = 1, \ \lambda_2 = -1$$

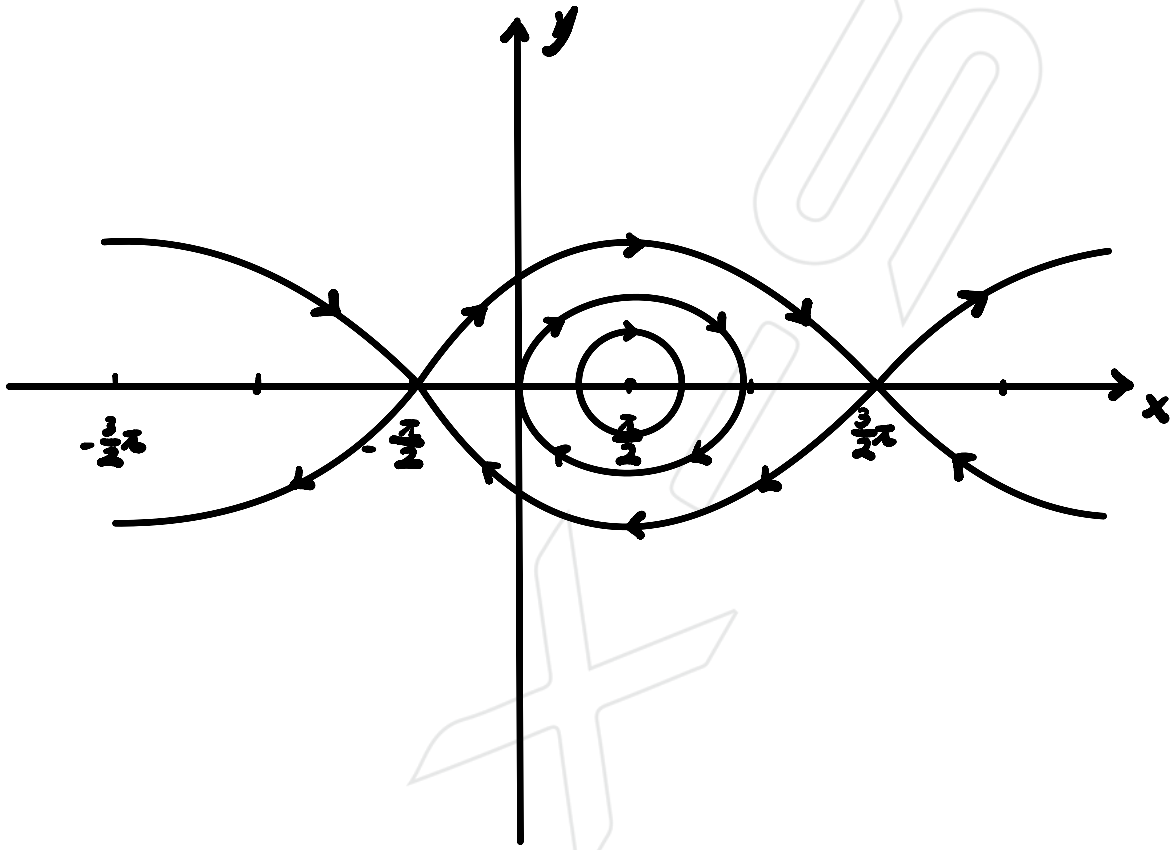
for $\lambda_1 = 1$,

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

for $\lambda_2 = -1$,

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\therefore$



6. Linearise the system

$$\dot{x}_1 = (a - bx_2)x_1$$

$$\dot{x}_2 = (cx_1 - d)x_2$$

about each of its equilibrium points and determine its local state-space structure around them. Hence deduce the global state-space structure for $a = b = c = d = 1$.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} (a-bx_2)x_1 \\ (cx_1-d)x_2 \end{bmatrix}$$

$$\therefore J_{(x,\delta)} = \begin{bmatrix} a-bx_2 & -bx_1 \\ cx_2 & cx_1-d \end{bmatrix}$$

At singular point, $\dot{x}_0 = f(x_0, \delta) = 0$

$$\therefore \begin{cases} (a-bx_2)x_1 = 0 \\ (cx_1-d)x_2 = 0 \end{cases}$$

$$\therefore x_0 = \begin{bmatrix} x_{10} \\ x_{20} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} \frac{d}{c} \\ \frac{a}{b} \end{bmatrix} \quad (b, c \neq 0)$$

$$\text{or } \begin{bmatrix} x_1 \\ 0 \end{bmatrix} \quad (a = 0)$$

$$\text{or } \begin{bmatrix} 0 \\ x_2 \end{bmatrix} \quad (d = 0)$$

$$\therefore J_{x_0} = \begin{bmatrix} a & 0 \\ 0 & -d \end{bmatrix} \quad \text{at} \quad x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore -(a-\lambda)(d+\lambda) = 0 \qquad \therefore \lambda_{1,2} = a \text{ and } -d$$

$$\therefore J_{x_0} = \begin{bmatrix} 0 & -\frac{bd}{c} \\ \frac{ac}{b} & 0 \end{bmatrix} \quad \text{at} \quad x_0 = \begin{bmatrix} \frac{d}{c} \\ \frac{a}{b} \end{bmatrix} (b, c \neq 0)$$

$$\therefore \lambda^2 + ad = 0 \qquad \therefore \lambda = \pm\sqrt{-ad}$$

$$\therefore J_{x_0} = \begin{bmatrix} 0 & -bx_1 \\ 0 & cx_1 - d \end{bmatrix} \quad \text{at} \quad x_0 = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} (a = 0)$$

$$\therefore (cx_1 - d - \lambda)(-\lambda) = 0$$

$$\therefore \lambda_1 = cx_1 - d, \ \lambda_2 = 0$$

$$\therefore J_{x_0} = \begin{bmatrix} a - bx_2 & 0 \\ cx_2 & 0 \end{bmatrix} \quad \text{at} \quad x_0 = \begin{bmatrix} 0 \\ x_2 \end{bmatrix} (d = 0)$$

$$\therefore (a - bx_2 - \lambda)(-\lambda) = 0$$

$$\therefore \lambda_1 = a - bx_2, \ \lambda_2 = 0$$

for $a = b = c = d = 1$,

when $x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $J_{(x_0, s)} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$$\therefore \lambda_1 = 1 \qquad \therefore \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\therefore \lambda_2 = -1 \qquad \therefore \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

when $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $J_{(x_0, \delta)} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$\therefore \lambda_1 = i \qquad \therefore \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \xi_1 = 0 \qquad \therefore \xi_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\therefore \lambda_2 = -i \qquad \therefore \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \xi_2 = 0 \qquad \therefore \xi_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$